

Verses, Valence, and Vectors: NLP Approaches to Musical Emotion

Abstract

Predicting a song's emotional tone only using its lyrics is challenging due to informal language and subversive context that often contradicts musical mood. This study trains and tunes popular encoder/decoder transformer models (BERT and RoBERTa) for continuous valence prediction (0-1), comparing them against Bag-of-Words baselines and Spotify's audio features. As expected, BoW representations fail, while BERT improves but remains negative ($R^2 = -0.04$ post fine-tuning). RoBERTa, pre-trained on 58 million tweets, achieves stronger results due to exposure to informal, emotionally expressive text. A combined model integrating fine-tuned RoBERTa with audio features achieves $R^2 = 0.233$, the highest of any model tested.

These findings demonstrate that lyrical analysis can match audio-based valence prediction when models are pre-trained on linguistically similar corpora and affirms the potential for multimodal MIR approaches where text and audio provide complementary signals.

Introduction

Music is a universal medium for emotional expression, yet the computational understanding of what makes a song feel happy or sad remains an open challenge. In Musical Information Retrieval (MIR), the field that applies computational methods to music analysis, sentiment is often represented through valence, a continuous measure from 0 to 1 that reflects the perceived musical positiveness of a track. High-valence songs typically sound cheerful or euphoric, whereas low-valence music conveys sadness, tension, or anger.

Traditionally, valence is estimated using audio-signal features. However, I was interested in whether it could also be inferred, at least partially, from a song's lyrics. Lyrics carry dense semantic content, and certain words ("love," "sunshine," "summertime") strongly evoke positivity, while others ("heartbreak," "tears," "fury") suggest negativity. Yet the relationship between lyrical sentiment and perceived musical emotion is often misaligned. I refer to this as the "Born in the U.S.A. paradox": cases where a song's musical valence sharply diverges from the emotional tone of its lyrics. These mismatches represent, in my view, the most intriguing aspect of the problem and challenge assumptions baked into leveraging models trained on fixed grammars to perform NLP-based sentiment analysis of lyrics.

This paper evaluates the performance of several multimodal models designed to predict valence using both lyrical sentiment and audio features. To directly examine the “Born in the U.S.A. paradox,” I conduct ablation studies to determine the extent to which audio features alone, lyrical sentiment alone, or their combination contribute to accurate valence prediction.

Background

2.1 Audio-based Music Information Retrieval

In 1980, Russell et al. demonstrated that musical emotion could be described along two axes: valence (pleasure–displeasure) and arousal (activation–deactivation)² and derived by an analysis of audio components such as harmony, tempo, and energy. This study builds on that foundation by combining audio features with language-based analysis of song lyrics retrieved through the Genius API. Specifically, I use a dataset of roughly 30,000 tracks, along with 12 audio features obtained through the Spotify API and published on Kaggle, as the foundation for my analysis.

2.2 Transformer-based lyrical analysis

Transformer architectures have fundamentally reshaped sentiment analysis. By leveraging attention mechanisms and massive training corpora, these models excel at modeling syntax, capturing contextual dependencies, and identifying subtleties in text. Yet, despite these advances, the benefits are largely unrealized in MIR, where lyric sentiment analysis remains a persistent challenge. Early large-scale studies found that lyric-only features delivered modest (if any) mood classification performance, with results varying widely across songs, moods, and genres³. Later work that incorporated richer semantic and structural features in lyric analysis showed modest performance improvements⁴ and highlighted the potential for improved performance, particularly when models were fine-tuned to the task. Given these mixed but incremental gains, applying modern transformer-based sentiment models to lyrics remains an inconsistently effective, but promising strategy.

Methods

As this is ultimately a regression task, I evaluate each model’s accuracy using R^2 and their errors using both root mean squared error (RMSE) and mean absolute error (MAE). Using these metrics comparisons clarify the extent to which advanced models improve multimodal MIR sentiment prediction by detecting subversive language, applying unconventional grammars, and accurately mapping lyrical content to emotional valence.

1. **[baseline]** Bag-of-Words / TF-IDF: Traditional count-based representations serve as non-contextual baselines to predict a phrase’s sentiment. This approach is included to quantify the improvement gained from the adoption of more sophisticated transformer embeddings.

2. BERT: A transformer model pre-trained on large-scale English text from Wikipedia and BookCorpus. It was selected for this task due to its strong ability to capture contextual word meaning and syntactic structure, especially in lengthy passages.
3. RoBERTa⁴: A model fine-tuned on ~58 million tweets for sentiment classification. Selected due to its flexibility in interpreting informal, colloquial text and non-standard grammar.

Results and Discussion

Baseline Results: Bag-of-Words Approach (ref: 3_Baseline_BoW_model.ipynb)

	Audio Only	Lyrics Only	Combined
R^2	0.2320	-17.3491	-2.1558
RMSE	0.1886	0.9220	0.3824
MAE	0.1533	0.6864	0.2911

The audio-only baseline establishes a strong benchmark for valence prediction, achieving an R^2 of 0.23. This result underscores the power of domain-specific acoustic signals like energy, danceability, and tempo, which were explicitly designed to capture musical mood. In contrast, the failure of the lyrics-only BoW model ($R^2 = -17.35$) confirms that word frequency counts lack the linguistic nuance required for sentiment analysis, and re-establishes the need for more sophisticated modeling. Along those lines, the naïve Bag-of-Words representation of lyrics significantly degraded performance when combined with audio features (R^2 dropping from 0.23 to -2.16).

BERT results (ref: 4_BERT_model.ipynb)

BERT was a compelling choice for the first transformer-based iteration of this exercise, given its ability to capture syntactic dependencies, its flexible architecture for downstream fine-tuning, and its extensive pre-training on 3.3 billion words of written English text.

	Lyrics Only	Combined	Lyrics Only <i>Post- fine tuning</i>
R^2	-0.2344	-0.1357	-0.0394
RMSE	0.2339	0.2243	0.2146
MAE	0.1909	0.1824	0.1732

BERT demonstrates substantial improvement over the Bag-of-Words baseline. The frozen lyrics-only model achieves $R^2 = -0.23$, a dramatic recovery from the BoW's $R^2 = -17.35$, in addition to reducing RMSE from 0.92 to 0.23 and MAE from 0.69 to 0.19. This, paired with similar improvements in the combined model, suggests that unlike BoW, BERT representations can complement audio more capably.

However, two fundamental limitations constrain BERT's off-the-shelf performance on this task.

1. **BERT's training corpus** (Wikipedia and BookCorpus) contains formal, structured prose that bears little resemblance to song lyrics. I address this in the next section by implementing RoBERTa.
2. BERT's architecture was **fundamentally designed for discrete label prediction**. The [CLS] token representation is optimized for distinguishing between classes, not for capturing the continuous emotional gradient that valence represents. That said, we can fine tune BERT by implementing a classification head designed to handle continuous variables.

Fine-tuning yields additional improvement, pushing the lyrics-only model from $R^2 = -0.23$ to $R^2 = -0.04$ —approaching the predictive baseline of the mean. This progression demonstrates that task-specific adaptation helps BERT learn aspects of the lyrics-valence relationship, though the signal remains fairly weak.

RoBERTa results (ref: 5_RoBERTa_model.ipynb, 6_Combined_ft_RoBERTa.ipynb)

In class, we discussed RoBERTa as an alternative to BERT, with the added advantage of a different pretraining corpus composed of the sort of nonstandard, subversive language commonly found in song lyrics - the results of this attempt are shown in the below table.

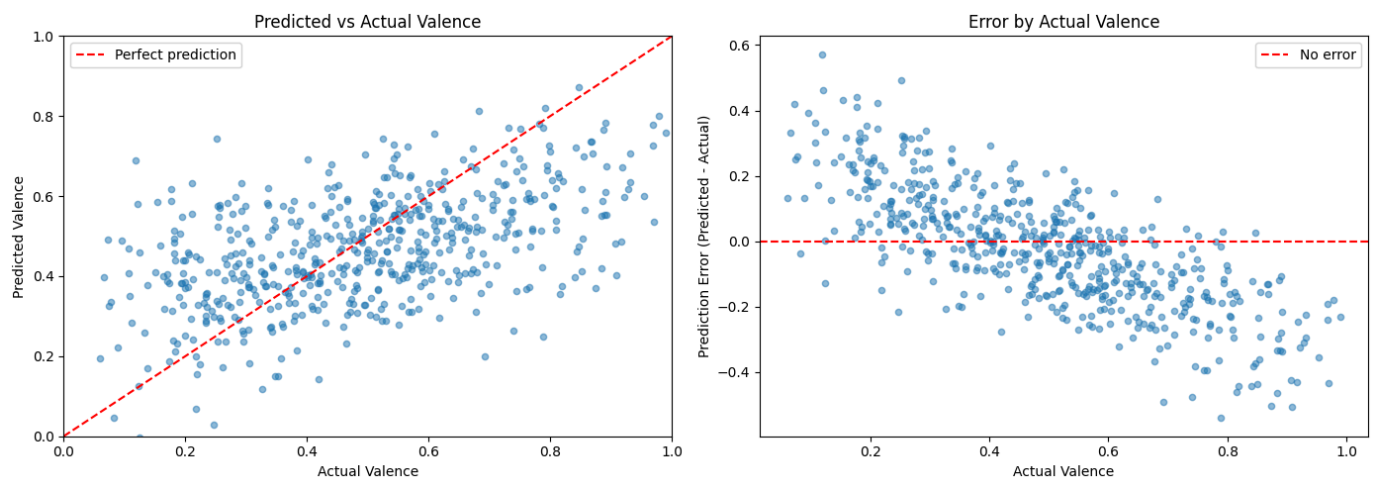
	Lyrics Only	Combined	Lyrics Only <i>Post-fine tuning</i>	Combined <i>Post-fine tuning</i>
R^2	0.0571	0.1335	0.0979	0.2334
RMSE	0.2044	0.1959	0.1999	0.1842
MAE	0.1666	0.1569	0.1634	0.1458

The results reveal a striking pattern of improvement. Even the frozen lyrics-only RoBERTa model achieves a positive R^2 (0.057), a marked departure from BERT's negative values, making it the first lyric model in this paper to outperform the mean baseline. This suggests that RoBERTa's tweet-based training provides representations that are inherently more aligned with lyrical sentiment than BERT's. Once again, the combined frozen model ($R^2 = 0.134$) demonstrates that RoBERTa embeddings complement audio features more effectively than BERT, likely because the model has learned to interpret emotional language that audio features alone cannot

capture—sarcasm, slang, and figurative expressions that are common in both tweets and song lyrics.

Finally, and most interestingly, the combined fine-tuned model achieves $R^2 = 0.233$, matching the audio-only baseline ($R^2 = 0.232$). This is a significant result: it demonstrates that lyrical analysis, when implemented with an appropriately pre-trained transformer and task-specific fine-tuning, can match the predictive power of audio features. This finding affirms the potential for text analysis to supplement audio features in MIR sentiment prediction, provided models possess the ability to reason non-linearly, project continuous variables, and leverage domain-relevant corpora. Prior to the conclusion, I take some time to

Discussion of Errors



Error analysis reveals even dispersion across the valence spectrum, with a characteristic "reversion to the mean" pattern where extreme values are mis-predicted at a higher rate. The worst predictions occur for songs with contrasting lyrical and musical sentiment - precisely the subversive cases that motivated this research. A sample of 10 such songs, their features & their lyrics are available in the appendix.

Conclusion

This study evaluated the application of transformer-based NLP models to musical valence prediction using song lyrics, comparing their performance against traditional text representations and an array of audio features.

- Beginning with a Bag-of-Words baseline ($R^2 = -17.35$), I demonstrated the inadequacy of non-contextual text representations for this task.
- BERT improved substantially ($R^2 = -0.23$ frozen, -0.04 fine-tuned), confirming that contextualized embeddings capture meaningful semantic relationships, though its formal training corpus limited its effectiveness on informal lyrical text.

- RoBERTa, pre-trained on tweets and thus exposed to colloquial, emotionally expressive language, achieved the strongest results. The combined fine-tuned RoBERTa model ($R^2 = 0.233$) matched the audio-only baseline ($R^2 = 0.232$), demonstrating that lyrical analysis can achieve parity with acoustic features when models are appropriately fine-tuned for continuous prediction.
- Error analysis revealed balanced performance across the valence spectrum, with the expected regression-to-mean pattern and failures concentrated on songs with contrasting lyrical and musical sentiment—the subversive cases that define this challenge.

These findings affirm that text-based valence prediction can supplement, and potentially enhance, audio-based approaches in Music Information Retrieval. The key insight is corpus alignment: models trained on formal text struggle with lyrics, while those exposed to informal, emotionally rich language transfer more effectively.

Two extensions would strengthen this research

1. Training a transformer variant on a corpus of semantically complex, like poetry, song lyrics, or literary fiction, could interpret the nuance of lyrics more effectively
2. Feature importance analysis of the audio components would clarify which acoustic signals (e.g., energy, tempo, mode) contribute most to valence prediction, informing more targeted multimodal architectures and attention-like weighting of inputs.

Together, these directions could further close the gap between what we hear and what we read in understanding musical emotion.

Appendix - Error table

For your academic edification & listening pleasure, here's a playlist with these songs:

<https://open.spotify.com/playlist/6VtXVC6NDG5MWnUDzvZ2my?si=6244cf0c691c49e8>

track_name	track_artist	actual_valence	predicted_valence	error	lyrics_preview
Take Ü There (feat. Kiesza)	Jack Ü	0.118	0.69	0.572	link
Break on Through (To the Other Side)	The Doors	0.789	0.248	-0.541	link
Love Will Tear Us Apart	Joy Division	0.909	0.403	-0.506	link
Who Needs Love	Trippie Redd	0.874	0.369	-0.505	link
Rosa Parks	OutKast	0.251	0.744	0.493	link
Issues/Hold On	Teyana Taylor	0.692	0.2	-0.492	link

Different State of Mind	Kid Bloom	0.74	0.263	-0.477	link
Love Is A Battlefield	Pat Benatar	0.884	0.421	-0.463	link
Shower	Becky G	0.121	0.582	0.461	link
Til I Don't	Mokita	0.816	0.356	-0.46	link

References

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